

Necessary commands

notepad Dockerfile

`mv Dockerfile.txt Dockerfile`

`cat Dockerfile`

`git add .`

`git commit -m "index.html v1 dockerfile added"`

`git remote add origin https://github.com/maurya-Priti/devops-assignment4.git`

`git branch -M main`

`git push origin main`

`mkdir 8033_8038`

`cd 8033_8038`

`git clone https://github.com/maurya-Priti/devops-assignment4`

`cd devops-assignment4`

`ls`

Give all the commands in the pdf docker ones, kubectl ones and mini kube

Stopped thinking

From the PDF, the Docker, Kubernetes (kubect1), and Minikube commands are:

Docker

```
sudo docker build -t pritimaurya/devops-assignment4:v1 .
```

```
docker push pritimaurya/devops-assignment4:v1
```

Kubernetes (kubect1)

```
kubect1 create deployment devops-assignment-app1 --image=pritimaurya/devops-assignment4:v1 --port=8080
```

```
kubect1 expose deployment devops-assignment-app1 --type=NodePort --port=8080 --target-port=80
```

Minikube

```
minikube service devops-assignment-app1
```

These are the Docker, kubect1, and Minikube commands shown in the PDF.

Give all in one and for username in docker use divyamull

In GitHub use divyasai-svg

```
# Build Docker image
```

```
docker build -t divyamull/devops-assignment4:v1 .
```

```
# Push image to Docker Hub
```

```
docker login
```

```
docker push divyamull/devops-assignment4:v1
```

Create Kubernetes deployment

```
kubectl create deployment devops-assignment-app1 --image=divyamull/devops-assignment4:v1 --port=8080
```

Expose deployment as a NodePort service

```
kubectl expose deployment devops-assignment-app1 --type=NodePort --port=8080 --target-port=80
```

Open the application in Minikube

```
minikube service devops-assignment-app1
```

Update image after modifying index.html (if required)

```
docker build -t divyamull/devops-assignment4:v2 .
```

```
docker push divyamull/devops-assignment4:v2
```

```
kubectl set image deployment/devops-assignment-app1 \
```

```
devops-assignment-app1=divyamull/devops-assignment4:v2
```

GitHub commands (using your GitHub username)

```
git remote add origin https://github.com/divyasai-svg/devops-assignment4.git
```

```
git branch -M main
```

```
git push -u origin main
```

Give all git commands also

Create project directory

```
mkdir devops-assignment4
```

```
cd devops-assignment4
```

```
# Initialize Git repository
```

```
git init
```

```
# Create index.html
```

```
touch index.html
```

```
# Create Dockerfile
```

```
touch Dockerfile
```

```
# Check repository status
```

```
git status
```

```
# Add files
```

```
git add .
```

```
# Commit files
```

```
git commit -m "Added index.html and Dockerfile"
```

```
# Rename branch to main
```

```
git branch -M main
```

```
# Add GitHub remote
```

```
git remote add origin https://github.com/divyasai-svg/devops-assignment4.git
```

```
# Push to GitHub
```

```
git push -u origin main
```

```
# Clone the repository (on another system)
```

```
git clone https://github.com/divyasai-svg/devops-assignment4.git
```

```
# Go to project directory
```

```
cd devops-assignment4
```

```
# Check files
```

```
ls
```

```
# Build Docker image
```

```
docker build -t divyamull/devops-assignment4:v1 .
```

```
# Login to Docker Hub
```

```
docker login
```

Push image to Docker Hub

```
docker push divyamull/devops-assignment4:v1
```

#Run docker image

```
docker run -d -p 8080:80 --name devops-assignment4 divyamull/devops-assignment4:v1
```

Check running containers

```
docker ps
```

Stop the container

```
docker stop devops-assignment4
```

Remove the container

```
docker rm devops-assignment4
```

Create Kubernetes deployment

```
kubectl create deployment devops-assignment-app1 --image=divyamull/devops-assignment4:v1 --port=8080
```

Expose deployment

```
kubectl expose deployment devops-assignment-app1 --type=NodePort --port=8080 --target-port=80
```

Open service in Minikube

```
minikube service devops-assignment-app1
```

Check deployment

kubectl **get** deployments

Check services

kubectl **get** services

Check pods

kubectl **get** pods

Nginx and httpd base code

FROM httpd:latest

COPY . /usr/local/apache2/htdocs/

FROM nginx:latest

COPY . /usr/share/nginx/html/

Flask dockerfile base code

FROM python:3.10

WORKDIR /app

COPY . /app

EXPOSE 5000

```
CMD ["python", "app.py"]
```

Jenkins- GitHub script

```
pipeline {  
  
    agent any  
  
    environment {  
  
        DOCKER_CRED = credentials('Docker')  
  
    }  
  
    stages {  
  
        stage('fetch-code') {  
  
            steps {  
  
                git branch: 'main', url: 'https://github.com/divyasai-svg/assign4.git'  
  
            }  
  
        }  
  
        stage('build-image') {  
  
            steps {  
  
                sh 'docker build -t divyamull/add4:${BUILD_NUMBER} .'  
  
            }  
  
        }  
  
        stage('push-image') {  
  
            steps {  
  
                sh ' echo $DOCKER_CRED_PSW | docker login -u $DOCKER_CRED_USR --password-stdin'
```

```
    sh 'docker push divyamull/add4:${BUILD_NUMBER}'

  }

}

stage('update-minikube') {

  steps {

    sh 'kubectl set image deployment/dd4 add4=divyamull/add4:${BUILD_NUMBER}'

  }

}

}
```

To run Kafka

Terminal 1

zookeeper-server-start.sh config/zookeeper.properties

Terminal 2

kafka-server-start.sh config/server.properties

Terminal 3

kafka-topics.sh --create --topic titanic --bootstrap-server localhost:9092

Terminal 4

Run producer and consumer codes

